UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 (5 marks)

Give ONE (1) appropriate example for each component in Von Neumann architecture.

QUESTION A2 (5 marks)

a) Name the inventor of the integrated circuit.

b) In which computer generation was the integrated circuit first introduced?

c) List THREE (3) main components of an integrated circuit.

QUESTION A3 (5 marks)

What is the decimal value of the sum of the binary numbers 1101 1100 and 0010 0111 if they are represented in sign and magnitude? Assume that the basic unit is the byte. Show your workflow.

QUESTION A4 (5 marks)

a) Discuss the computer interface that is available during the third and fourth computer generations.

b) In your opinion, what is the leading cause that changed the computer interface in the third to the fourth computer generations?

QUESTION A5 (10 marks)

Question 7 is based on the following scenarios.

While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks.

Hafiz was tasked to research on the history of Malay culture by his professor. He went to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk.

Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance.

Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a) Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

b) Describe SRAM and DRAM.

c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

QUESTION A6 (10 marks)

Question 10 is based on Figure 1. Figure 1 depicts a Non-uniform Memory Access (NUMA)

architecture.

a) Redraw a symmetric multiprocessor organization based on Figure 1.

b) Give TWO (2) examples of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

c) Discuss TWO (2) disadvantages of tightly coupled Multiple Instruction, Multiple Data (MIMD) parallel processor architectures.

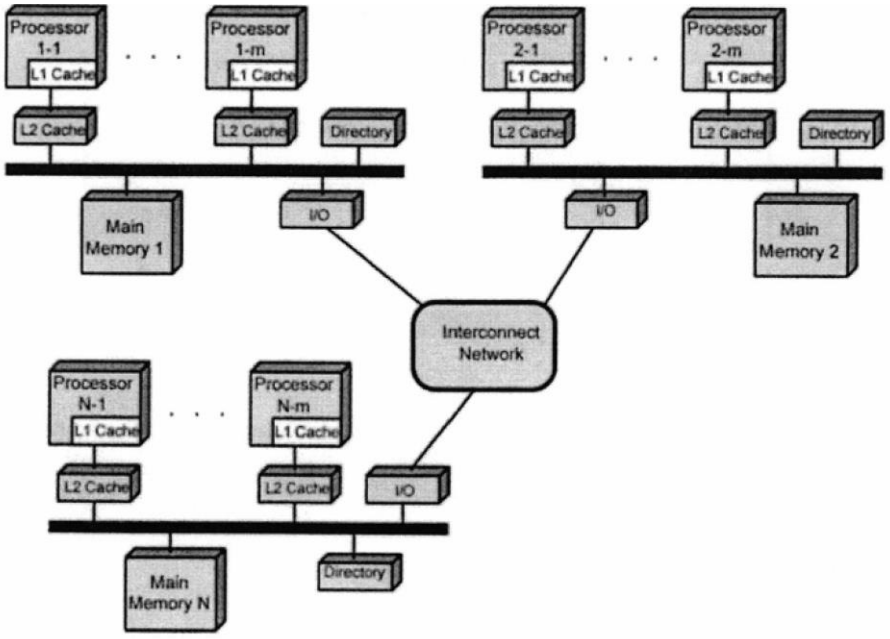


Figure 1: Non-uniform 'Memory Access (NUMA)

QUESTION A7 (10 marks)

a) Discuss the TWO (2) processes in the machine cycle that are governed by the main memory.

b) List TWO (2) operations that are comprised in the Execute phase.

c) Explain what happens in the Program Counter (PC) and Instruction Register (IR) during the Fetch cycle.

QUESTION A8 (10 marks)

Describe the specific function of each registers in the CPU.

QUESTION A9 (10 marks)

Question 1 is based on the following statement.

In 1942, Mr. X and Mr. Y proposed an all-electronic calculating machine. The U.S. Army, meanwhile, needed to calculate complex wartime ballistics tables. The result was Electronic Numerical Integrator And Computer (ENIAC) that was built between 1943 and 1945. ENIAC is the first large-scale computer to run at electronic speed without being slowed by any mechanical parts. For a decade, until a 1955 lightning strike, ENIAC may have run more calculations than all mankind had done up to that point.

a) In which computer generation, ENIAC is developed?

b) Who built the ENIAC machine?

c) What is the main technology used in the ENIAC machine?

d) Outline a possible solution for each of the following problems that are faced when ENIAC is used.

Problem	Solution
Produced a lot of heat	
Consume a lot of ener	
Lar e in size	
Slow rocessin	
Limited ro rammin ca abilities	

QUESTION A10 (10 marks)

a) Convert -176.812510 to floating point number in IEEE single precision format in hexadecimal value form.

b) Convert 3FOOC00016 in IEEE single precision floating point number to decimal value form.

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 (20 marks)

Question 1 is based on Figure 2 and Table 1.

Write the appropriate programs using correct assembly language notations and comments to compute the equation represented in Figure 2 using zero-, one-, and three-address instructions. The instructions available for use are provided in Table 1. Note that Register A, B, C, and D cannot be modified, and Register x, y, and z can be used as temporary registers, However, the final answer must be stored in Register y.

$$y = \frac{A^2 - (B + AC)}{AD}$$

Figure 2

Table 1: Machine Instruction

Zero-address	One-address	Three-address
PUSH	LOAD	ADD
pop	STOR	SUB
ADD	ADD	MUL
SUB	SUB	DIV
MUL		
DIV	DIV	